R REGRET

ALPACA AI TRADING AGENTS HACKATHON 2026

AUTONOMOUS AI QUANTITATIVE SYSTEM

Defined-Risk Options Trading With Zero Tail Risk

The next-generation trading agent combining **Featherless AI qualitative market reasoning** with **deterministic Python mathematical risk bounds**.

$100,000	6 GATES	70-80%	0.0%
ALPACA PAPER BASELINE	DETERMINISTIC SAFETY	TARGET SPREAD WIN RATE	NAKED LOSS EXPOSURE

Bridging AI Intelligence with Wall Street Guardrails

Modern traders face a dilemma: LLMs possess incredible macroeconomic synthesis and context parsing, yet they are mathematically brittle when calculating options Greeks and managing downside risk.

REGRET solves this by creating a hardened software barrier: AI generates the qualitative strategic thesis, while deterministic Python gates enforce 100% of the trade execution, Greek boundaries, and position limits.

MISSION

To make quantitative, defined-risk options strategies autonomous, explainable, and mathematically protected against black-swan liquidation.

TARGET EDGE

Systematically harvesting elevated Implied Volatility overestimation and positive Theta time decay on high-volume indices and equities ($SPY, $QQQ, $NVDA).

The 3 Critical Flaws in AI Trading Bots

Why conventional language model trading bots destroy Sharpe ratios and fail institutional risk audits.

📐 Hallucinated Math

LLMs cannot reliably calculate real-time options Greeks (Δ, Γ, Θ) or margin collateral inside prompt space, generating catastrophic sizing errors.

💥 Naked Tail Risk

Unconstrained agents execute naked short options without purchasing outer wings, creating unlimited loss liability on single-day market volatility surges.

🔒 Black Box Fragility

End-to-end neural network bots provide zero explainability. When market regimes shift abruptly, unmonitored bots freeze or panic-sell at maximum drawdown.

How We Solve It

We created a strict separation of concerns that routes market intelligence through four hardened stages:

01 Alpaca Market Data Screener
Streams real-time stock bars & option chains. Computes 52-week IV Rank.

02 Featherless AI Strategic Synthesis
Qwen 2.5 72B & Llama 3.3 evaluate volatility skew and market regime bias.

03 6 Deterministic Hard Risk Gates
Validates max loss ($500), daily halt ($2k), Theta > 0, and bid-ask spread.

04 Alpaca Multi-Leg (mleg) Order Router
Submits native atomic credit spreads directly to paper-api.alpaca.markets.

regret_autonomous_agent.py

```python
# 1. Run live market screener
metrics = screener.calculate_iv_rank("SPY")
if metrics.iv_rank ≥ 35:
    # 2. Featherless AI Synthesis
    thesis = llm.generate_strategy_thesis(metrics)
    # 3. 6 Hard Python Risk Gates
    passed = risk_engine.validate(thesis)
    if passed.approved:
        # 4. Multi-leg Spread Routing
        broker.submit_spread_order(thesis.orders)

>> [OK] Executed 5 defined-risk spreads on PA3XUIGQ0VGB
```

The 6 Hard Risk Gates

Every trade proposal must clear 100% of these hard-coded checks. Failure on any gate halts execution immediately.

GATE 01
Max Loss / Trade
Hard ceiling of **$500 max loss** per trade. Oversized allocation is mathematically blocked.

GATE 02
Daily Circuit Breaker
Automatically halts all new trade entries if cumulative realized daily loss reaches **$2,000**.

GATE 03
Position Ceiling
Limits open positions to a maximum of **5 concurrent spreads** to eliminate concentration risk.

GATE 04
Bid-Ask Health
Rejects illiquid option contracts where the bid-ask width exceeds **10% of credit received**.

GATE 05
Greeks Sanity
Requires net positive Theta (**$\Theta > 0$**) and Delta strictly bounded within ±0.40.

GATE 06
Expiration Safety
Rejects contracts with **<7 DTE** at entry to eliminate weekend assignment & pin risk.

How We Capture Positive Expectancy

Implied Volatility consistently overstates realized volatility >80% of the time. We sell elevated premium and close systematically:

🎯 **50% Profit Target (Take Profit)**
Automatically closes spreads once 50% of maximum credit is captured, optimizing capital velocity.

🛑 **2.0× Stop Loss Gate**
Closes open spreads if unrealized loss reaches 2.0× credit, cutting adverse moves immediately.

⏱️ **1 DTE Pin Risk Exit**
Closes all open positions 24 hours prior to expiration to eliminate weekend assignment uncertainty.

Spread Payoff Profile

> 0	7-45
POSITIVE THETA	TARGET DTE WINDOW

Live Alpaca Execution & Traction

Account PA3XUIGQ0VGB actively trading with 10 covered legs across 5 defined-risk multi-leg spreads.

$100,000	5 SPREADS	10 LEGS	95 / 95
BASELINE BALANCE	MULTI-LEG EXECUTED	ACTIVE COVERED POSITIONS	TEST PASS RATE (100%)

SYMBOL	SETUP TYPE	SHORT LEG	LONG PROTECTION	MAX LOSS	ALPACA STATUS
SPY	Bear Call Spread	766.0 Call	767.0 Call	$15.00	FILLED (mleg)
QQQ	Bear Call Spread	710.0 Call	711.0 Call	$15.00	FILLED (mleg)
IWM	Bear Call Spread	294.0 Call	295.0 Call	$15.00	FILLED (mleg)
NVDA	Bear Call Spread	225.0 Call	227.5 Call	$165.00	FILLED (mleg)
AAPL	Bear Call Spread	325.0 Call	327.5 Call	$165.00	FILLED (mleg)

REGRET

Let's Build the Future of AI Trading

REGRET sets a new standard for autonomous trading agents: combining open-source LLM intelligence with mathematically unbreakable risk boundaries.

🚀 **Live Web App**
`https://regret.fly.dev`

⚡ **Autonomous CLI**
`regret agent start`

🏆 **Paper Account**
`PA3XUIGQ0VGB`